

Session 24 : Analyze & Synthesize

Today you examine AI in your domain through five lenses, polish everything, draft the remaining sections, and start your presentation



Today's Goals

By the end of this session, you'll have accomplished the following:

01

Five Lenses Applied

Examined AI in your domain through Bias, Safety, Ethics, Human-in-the-Loop, and Voice Integrity

02

Three Sections Drafted

Bias & Fairness, Safety & Governance, and Conclusion & Future Work

03

All Six Sections Polished

Every drafted section reviewed, sharpened, and in your authentic voice

04

Abstract & Attribution Written

Final Abstract and AI Attribution Statement complete

05

Presentation Started

First three Gamma slides built - homework finishes the rest

Today's work adds

New Paper Sections

Bias & Fairness Analysis

Safety & Governance

Conclusion & Future Work

Final Abstract

Replaces the placeholder from S22

Additional Deliverables

AI Attribution Statement

Transparent disclosure of AI tools used

Gamma Presentation

First three slides started today

- ✓ By end of homework after today, your paper is complete and your presentation is ready.

Homework Review



Flag Generated Phrasing

Sentences that sound polished but don't match your authentic voice will be identified and marked for revision.



Connect Surprises to Analysis

Your mentor will link your documented surprises to potential bias or safety considerations for today's work.



Push on Specificity

Places where claims could be more concrete and specific will be challenged - vague statements need evidence.



Identify Strongest Observations

The most powerful observations from your exploration will be flagged as anchors for the analysis ahead.



The Five Lenses

Your analysis today applies five lenses to AI applications in your domain. These aren't all separate paper sections - they're ways of looking at AI in your domain to surface what matters.

Lens	What It Examines
Bias & Fairness	Who AI in this domain works well for; who it fails or underserves
Safety & Governance	What harms could result; who's accountable
Ethics	What values are at stake; who benefits, who could be harmed
Human-in-the-Loop	Where human oversight exists or should exist
Voice Integrity	Whether your paper is in your authentic voice

❗ Bias & Fairness and Safety & Governance become their own paper sections. Ethics, Human-in-the-Loop, and Voice Integrity integrate into those sections.



Lens 1 - Bias & Fairness

You learned the bias frameworks in Session 20. Today you apply them to AI as it's actually being used in your domain right now - not to a model you built.

Data Bias

What's in and missing from the data AI systems are trained on

Algorithmic Bias

How the algorithm processes information and what patterns it amplifies

Human Bias

What assumptions the people building or using the system bring to it

Bias Questions to Ask About Your Domain

For your Bias & Fairness analysis, work through these questions. Your sources will already have raised some of them - your job is to synthesize and connect.

→ **Training Populations**

What populations are AI applications in this domain trained on? Who's overrepresented? Who's missing?

→ **Hands-On Patterns**

What did your hands-on exploration reveal about who or what the technology handles well versus poorly?

→ **Published Failures**

Where do the published failures or critiques of AI in this domain cluster? Whom does it consistently underserve?

→ **User Testing Signals**

What did your user testers notice about who the tool seemed to work for or not work for?

Bias - A Concrete Example

Food Allergy Management for Children

What Sources Revealed

Allergen-detection systems are typically trained on Western foods, which means they may underperform on foods central to non-Western cuisines.

What Exploration Confirmed

The classifier handled Western examples (peanut butter, Snickers) reliably but had no training data for Asian peanut sauces, African peanut stews, or Middle Eastern peanut-based sweets.

The Specific Claim

This data gap means the tool could fail for the populations whose families eat these foods — populations who are not less in need of allergen support, just less represented in training data.

This mirrors patterns documented in other AI systems, where narrow training data produces systems that work for some users and fail for others.

Lens 2 - Safety & Governance

You learned the safety frameworks in Session 21. Today you apply these three categories to your domain.



Misuse

How AI in this domain could be used to cause harm - intentionally or through negligence

Privacy

What personal data AI applications collect and how that data could be exposed or exploited

Accountability

Who's responsible when AI in this domain fails or causes harm - the developer, the user, the company?

Safety Questions to Ask About Your Domain

Work through these questions as you draft your Safety & Governance section. Connect your answers to your hands-on exploration and user testing wherever possible.



Realistic Misuse

What are realistic ways AI in this domain could be misused?



Sensitive Data

What sensitive data do these AI applications collect? How could it be exposed?



Accountability Gap

When something goes wrong, who's accountable? Is there a clear answer?



User Concerns

Did your hands-on exploration or user testers surface specific safety risks or concerns?

Safety - A Concrete Example

Food Allergy Management for Children

The most serious safety concern in this domain is **misplaced trust**. Sources reported that consumer-facing allergen detection tools often present results with high confidence even when wrong.

What Exploration Revealed

When the classifier misidentified a Snickers bar as peanut-free, it did so with high confidence - exactly the failure mode most likely to cause harm. A parent or child trusting that result could face a life-threatening allergic reaction.

The Accountability Problem

If a child consumes an allergen because they trusted a misclassification, responsibility is distributed across the parent, the tool maker, and the platform that hosts the model. There's no clear standard for AI accessibility tools serving vulnerable populations like children with allergies.

① Specific. Connects literature, hands-on exploration, and broader accountability concerns.

Lens 3 - Ethics

Ethics looks at the broader values at stake in AI applications in your domain. These observations integrate into your Bias and Safety sections — they don't get their own paper section, but they sharpen the analysis in both.



Who Benefits?

Who benefits from AI applications in this domain? Are those benefits distributed equitably?



Who Could Be Harmed?

Are there populations who can't say yes or no to how AI is being applied to them? Children, patients, elderly people, animals?



Values in Tension

Speed vs. accuracy. Privacy vs. functionality. Individual benefit vs. collective risk. What tensions exist in your domain?



Lens 4 - Human-in-the-Loop

AI works best when humans are in the loop not when AI replaces human judgment. This principle has been reinforced throughout the program. Now apply it to your domain.



Human-in-the-loop observations integrate into your Safety & Governance section as part of what responsible deployment looks like in your domain.



Lens 5 - Voice Integrity

This is the lens you apply to your own writing. Every sentence in your paper must be yours.

Not Paraphrased AI Output

Text you didn't really process or make your own doesn't belong in your paper

Not Copy-Pasted from a Source

Even well-cited material must be synthesized in your own reasoning

Your Voice, Your Reasoning, Your Judgment

Read your paper aloud. If a sentence doesn't sound like you when you say it, rewrite it.

⚠ If you can't say something in your own words, you don't understand it well enough yet. That's your signal to go back to the source.

AI-Sounding Phrases

Generated-sounding phrasing often shows up in predictable patterns. Learn to spot them - then rewrite.

Overly Polished Sentences

Don't match your own voice in earlier, more natural writing

Hollow Phrases

"In today's rapidly evolving landscape" or "leveraging cutting-edge technology"

Unusual Vocabulary

Words you wouldn't normally use in conversation or earlier writing

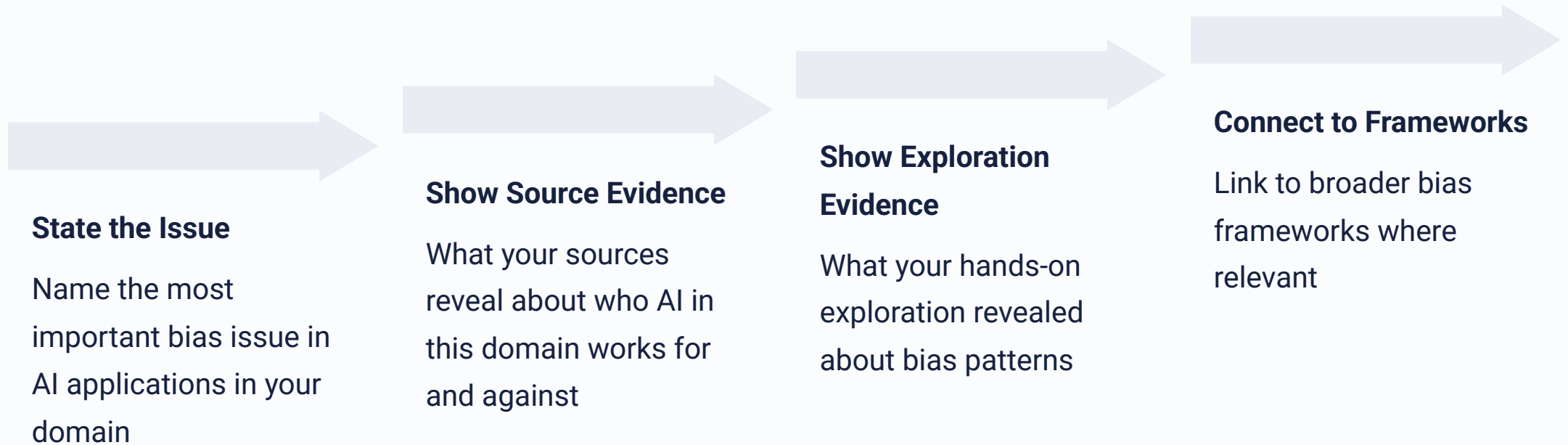
Generic Claims

No specific examples — sentences that could appear in any paper on any topic

① When you spot these, rewrite in your own words. Don't just swap one phrase for another - actually think through what you mean.

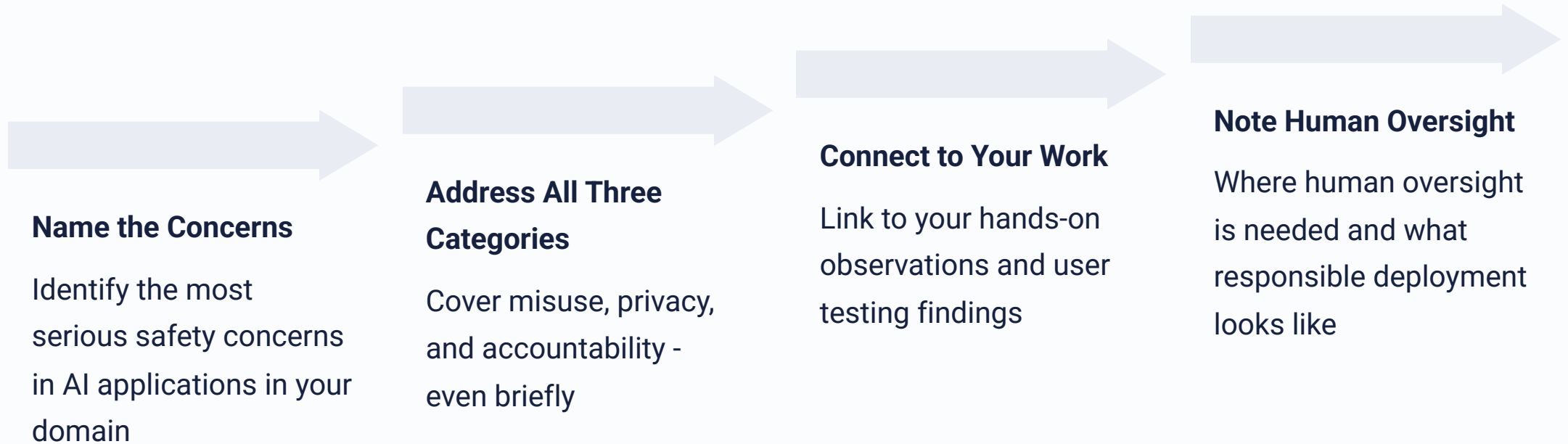
Drafting Bias & Fairness - Structure

Use this structure for your Bias & Fairness section. Follow the sequence, then check your target word count by grade.



Drafting Safety & Governance - Structure

Use this structure for your Safety & Governance section. Integrate Ethics and Human-in-the-Loop observations here - they don't get separate sections.



Polish

Polish doesn't mean making sentences fancier. It means making your paper sharper, clearer, and more authentically yours.

Remove Generated Phrasing

Replace AI-sounding language with your own voice

Make Claims Specific

Replace "many people" with "in the surveyed population of X"

Add Missing Citations

Every claim needs a citation or comes from your hands-on exploration

Improve Flow

Make sure each section sets up the next - check every transition

Cut Filler

Remove words that don't earn their place on the page

👉 A polished paper is **shorter, sharper, and more clearly in your voice** - not longer or more decorated.

Polish Order

When you polish, work through your sections in this specific order. Don't skip steps - each one catches different problems.

1

Read Aloud for Voice

Read each section aloud - if it doesn't sound like you, rewrite it

2

Check Every Claim

Every claim has a citation or comes from your hands-on exploration

3

Check Transitions

Does each section set up the next?

4

Cut Filler

Remove filler words and generic phrases that don't add meaning

 Don't polish your draft until it's perfect - that's a trap. Polish until it's clearly in your voice, well-cited, and flows. Then stop.

Drafting the Conclusion

Your Conclusion summarizes what your investigation showed about AI in your domain. It does not introduce new findings.

1 Restate the Investigation

What you investigated and why it mattered

2 Summarize Findings

What your sources and hands-on exploration revealed

3 State the Takeaway

The most important thing you learned about AI in your domain

4 Point Forward

Briefly: what comes next (this is Future Work)

 The Conclusion does not introduce new findings. Everything in it should connect back to something you already showed.

Writing the Final Abstract

Now you replace the placeholder Abstract from S22 with the real one. Use the same five-component formula - but with real findings instead of placeholders.

Component	Now You Can Fill In
Problem Statement	Why your domain investigation mattered
Research Question	Your sharpened question
Methodology Preview	How you investigated
Findings	What you actually found
Implications	What it means for your domain

 The Abstract is the last thing you write - but it appears first in your paper.

AI Attribution Statement

What It Is

A short statement disclosing which AI tools you used in your investigation and how you used them.

What It Isn't

It's not a confession. It's not an apology. It's **transparency** - being clear about what role AI played in your research process.

Why It Matters

This practice is becoming standard in academic and professional writing. Researchers, journalists, and scientists are increasingly expected to disclose AI assistance clearly and specifically.

By including an AI Attribution Statement, you're aligning with how serious researchers handle AI assistance and demonstrating the kind of intellectual honesty that defines Academic Integrity.

AI Attribution Statement - What to Include

Your statement should cover two things: what AI tools did, and what you did yourself.

For Each AI Tool You Used

Which tool

What you used it for

How you verified its output

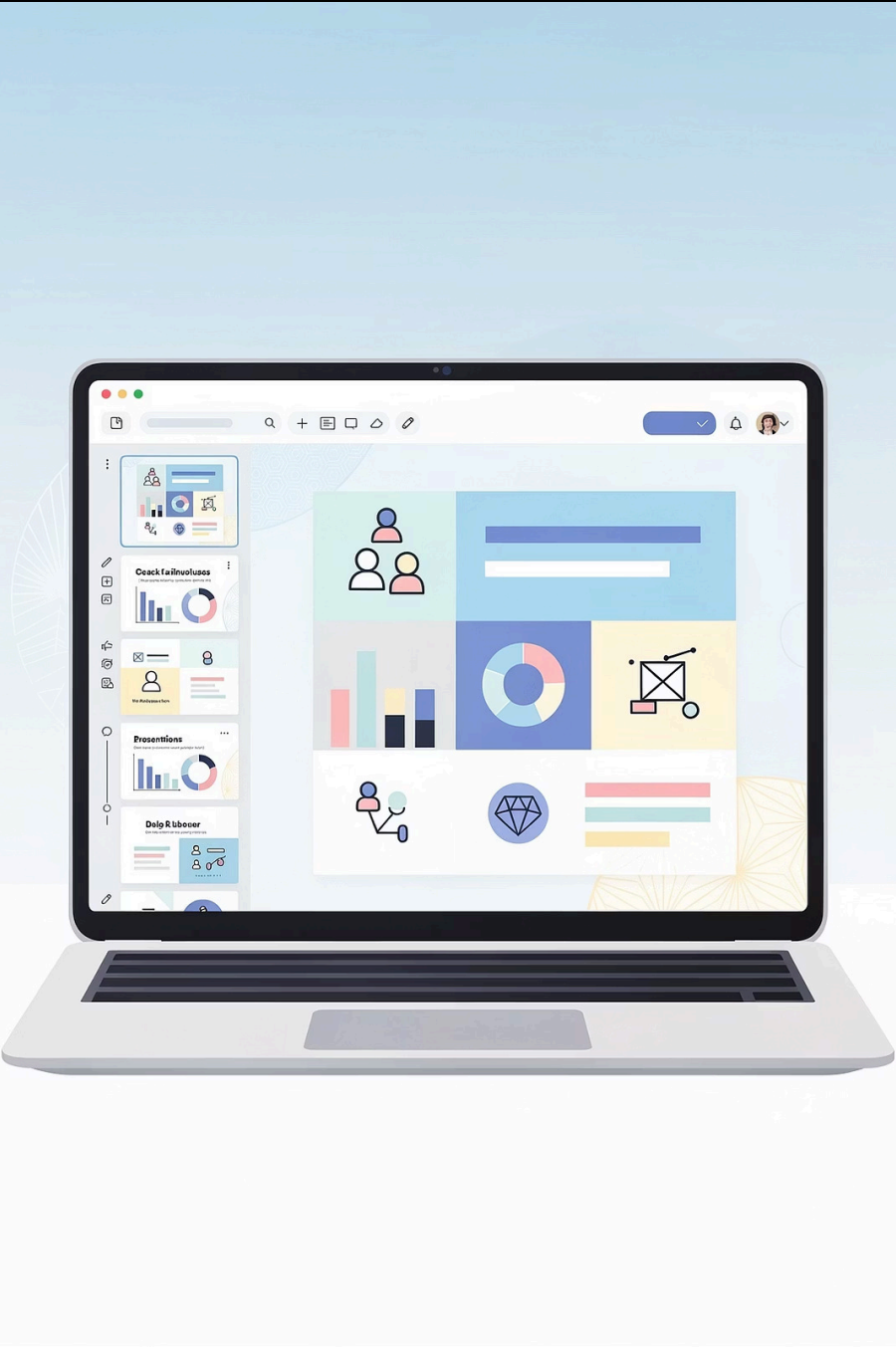
What You Did Yourself

- Drafting all sections
- Polishing in your voice
- Conducting the hands-on exploration
- Doing user testing
- Deciding what claims to make

AI Attribution Statement - Example

"AI tools assisted in this investigation. Perplexity helped discover candidate sources. NotebookLM was used to verify content and extract specific findings, with all extractions cross-checked against original sources. Gemini compared candidate sources during initial filtering. Teachable Machine was the subject of the hands-on exploration. ChatGPT was used briefly for early brainstorming on framing but not for content generation. All paper sections were drafted, polished, and revised by me. Where AI tools suggested phrasing, I rewrote in my own voice. My mentor provided guidance and feedback throughout but did not write any sections of this paper."

- ① This is one example. Yours should reflect what you *actually* used - the tools, the purposes, and the verification steps that match your real investigation process.



Meet Gamma

Now we move to your presentation. The tool you'll use is called Gamma.

What Gamma Is

An AI presentation tool that generates designed slides from text

How It Works

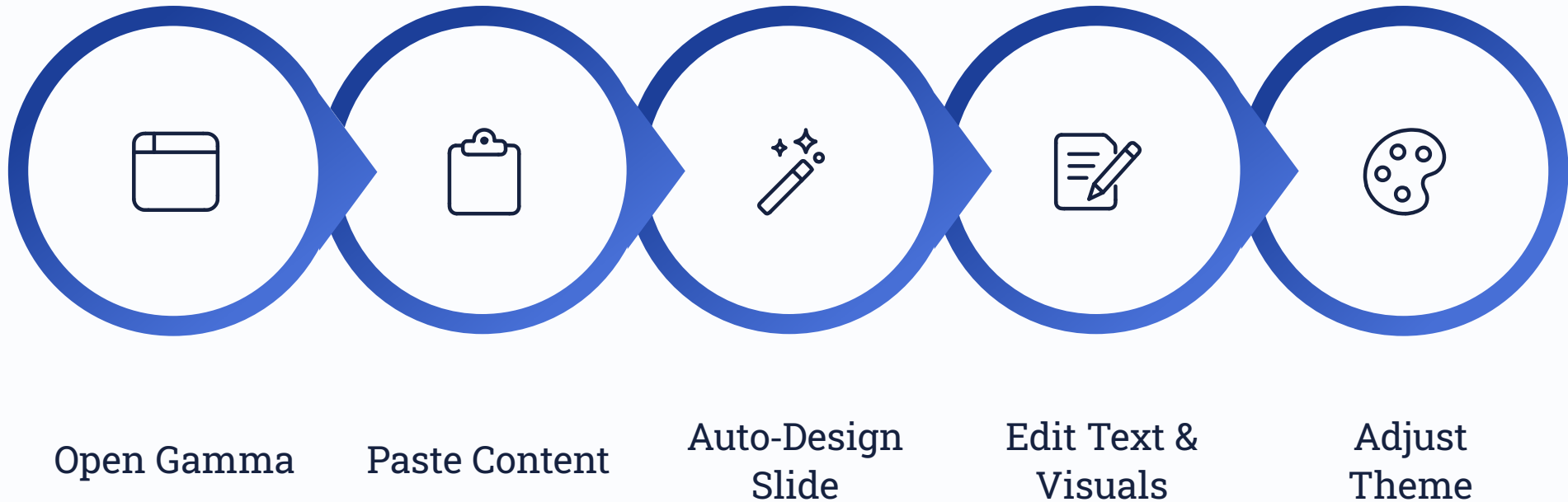
You provide the content; Gamma handles the design automatically

Why It's Powerful

Built-in slide design templates and visual style options - you focus on content, not layout

How Gamma Works

The basic Gamma workflow is straightforward. You don't have to think about layout or colors - you focus on content.



Gamma works because it has slide design templates and visual style options built in. Each step takes only a few minutes once you're familiar with the interface.

Your Presentation Structure

Your presentation will have nine slides. You'll start three today; six come in homework.

Slide	Title	Content from Your Paper
1	Title	Paper title + your name + grade
2	Hook	The most striking finding or question from your investigation
3	Research Question	Your sharpened research question
4	Background	Three key findings from your Related Work
5	Methodology	How you investigated (literature + hands-on)
6	Results	Your synthesis of findings
7	Bias & Safety	Main bias concern and main safety concern
8	Conclusion	Main takeaway and one or two future work directions
9	Thank You	Closing

Today's First Three Slides

In the time you have left in this session, you'll build your first three Gamma slides. Six more come in homework.

Slide 1 - Title

Your paper title, your name, your grade. Clean and clear — this is your first impression.

Slide 2 - Hook

One striking sentence that opens your presentation strongly. The most surprising or important thing you found.

Slide 3 - Research Question

Your sharpened research question, prominently displayed. This is what your whole investigation was built to answer.

 Six more slides come in homework. By Session 25, your full presentation is ready.

Outputs Checklist

By the end of today's session, you should be able to check off every item on this list.

Paper Sections

- ☐ Bias & Fairness section drafted
- ☐ Safety & Governance section drafted (with Ethics and HITL integrated)
- ☐ Conclusion & Future Work drafted
- ☐ All six prior sections polished

Additional Deliverables

- ☐ Final Abstract written
- ☐ AI Attribution Statement written
- ☐ References list complete and consistent
- ☐ First three Gamma slides started

Homework

Final Proofread

Citations consistent, voice consistent, no remaining generated-sounding phrases throughout the full paper

Build Slides 4-9

Complete the remaining six Gamma slides - Background, Methodology, Results, Bias & Safety, Conclusion, Thank You

Practice Aloud

Run through your full presentation aloud, time it, and aim for about 10-15 minutes

① By end of homework, your paper is complete and your presentation is ready for Session 25.

What's Next - Session 25

Bring everything to Session 25. You'll present your investigation, defend it, and reflect on the full journey.



Complete Paper

All 8 sections + AI
Attribution + References
- polished and in your
voice



Full Presentation

Your complete 9-slide
Gamma presentation,
ready to deliver



Practiced Delivery

A rehearsed
presentation you can
deliver confidently



Readiness to Defend

A willingness to defend
your work when your
mentor asks journal-
reviewer-style questions

Session 25 is **Deliver & Reflect**. You'll present your investigation, defend it under journal-reviewer-style questions, reflect on the 25-session journey, and receive your portfolio.